

CONTACT

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Paris, France

Age 25

clarence-pfister.github.io

github.com/Clarence-Pfister

LANGUAGES

Japanese
Native

French
Fluent — high school and higher education in France

English
Professional — TOEIC 840 (B2)

SKILLS

Programming
Python, C/C++, MATLAB, Git

Robotics software
ROS 2, Docker / Compose, Unitree SDK (DDS)

Perception
RGB-D & point clouds, semantic segmentation, OpenCV, OCR

Navigation
SLAM (slam_toolbox), A*, DWA, LiDAR

Learning & control
Reinforcement learning (PPO, RSL-RL), imitation learning (LeRobot), PID control, ONNX deployment

Simulation
Isaac Lab / Isaac Sim, MuJoCo, MJCF / URDF / USD models

Hardware
Unitree G1, Intel RealSense, ESP32, Arduino, SolidWorks

INTERESTS

Humanoid robotics, programming, audiovisual

Clarence PFISTER

Robotics Engineer — Humanoid Robots, Perception & Learning-Based Control

Engineering degree in Robotics, Polytech Sorbonne (end of November 2026) · Seeking a full-time position in Paris, available now

Robotics engineer with six months of hands-on work on the Unitree G1 humanoid, from simulation to the physical robot: reinforcement-learning environments in Isaac Lab, Sim2Sim validation in MuJoCo and deployment on hardware. Earlier experience in RGB-D perception (point clouds, semantic segmentation). Used to shipping working systems for industrial clients with ROS 2, Docker, Python and C++.

EXPERIENCE

Robotics Engineering Intern — final-year internship Apr – Oct 2026
KUFUSHA Co., Ltd. · Sagamihara, Japan

- **Humanoid driving automation (client project):** camera + OCR speed measurement and PID ankle control letting a G1 drive a vehicle simulator at a steady 60 km/h (±1 km/h); low-pass filtering removed pedal hammering. Demonstrated live at the client's technical review.
- **RL jumping (personal R&D):** custom Isaac Lab environment trained with PPO — six-phase reward machine, reference-state initialization, early terminations, per-joint action scaling. Diagnosed a conflict between tracking reward and termination penalty; the fix gave a policy that tracks the reference jump with zero falls and passes Sim2Sim validation in MuJoCo.
- **Sim-to-real (client choreography project):** domain randomization, observations restricted to real sensors, MuJoCo validation and ONNX export brought a learned policy onto the physical G1.
- **Robot model & motion data:** built a 23-DoF G1 model (MJCF + USD) matching the real robot's joints and limits; reference motions from text (KiMoDo) or video (GEM-X), retargeted with Holosoma.
- **Imitation learning:** set up a data-acquisition bench (PIKA gripper, 6-DoF hand tracking, LeRobot datasets).

Robotics Intern — Perception Jun – Aug 2025
KUFUSHA Co., Ltd. · Sagamihara, Japan

- Traversable-terrain detection and tall-grass height estimation from RGB-D data (Intel RealSense): point-cloud processing, semantic segmentation, CPU/GPU performance optimization, ROS 2 and Docker.

PROJECTS

Autonomous exploration & navigation — TurtleBot4 Nov 2025 – Feb 2026

- Pair project in ROS 2: mapping with SLAM (slam_toolbox); my part was the planning stack — A* global planner on the occupancy grid, DWA local planner with obstacles extracted from the LiDAR scan, return to the start pose.

Agricultural electric cart prototype Sep 2024 – Apr 2025

- Year-long open-source team project with a partner farm: ESP32-based control system, hardware/software integration.

Roguelike arena game in C++ (team of 3) Feb – Mar 2025

- Entity Component System (ECS) architecture, object-oriented and modular design.

EDUCATION

2023 – 2026 **Engineering degree (MSc level) in Robotics** — to be awarded end of November 2026
Polytech Sorbonne, Sorbonne University, Paris

2021 – 2023 **Preparatory classes for engineering schools** — Mathematics, Engineering Sciences
Lycée Raspail, Paris

2021 **French Baccalauréat**, with honors (Mention Bien) — Lycée Jacques Decour, Paris